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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Boolean Algebra

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

33	23	1990 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Answer key	at the end

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Marks 1 14 questions

Q1. GATE ECE 2024

For the Boolean function

$F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$,

the essential prime implicants are _____.

(A) $BD, \overline{B}\,\overline{D}$
(B) BD, AB
(C) $AB, \overline{B}\,\overline{D}$
(D) $BD, \overline{B}\,\overline{D}, AB$

Open solution

Q2. GATE ECE 2022

Select the Boolean function(s) equivalent to x + yz, where x, y, and z are Boolean variables, and + denotes logical OR operation.

(A) x + z + xy
(B) (x + y) (x + z)
(C) x + xy + yz
(D) x + xz + xy

Open solution

Q3. GATE ECE 2018

A function F(A, B, C) defined by three Boolean variables A, B and C when expressed as sum of products is given by

F = $\overline{A}.\overline{B}.\overline{C} + \overline{A}.B.\overline{C} + A.\overline{B}.\overline{C}$

where, $\overline{A}$, $\overline{B}$, and $\overline{C}$ are the complements of the respective variables. The product of sums (POS) form of the function F is

(A) F = (A + B + C)(A + $\overline{B}$ + C)($\overline{A}$ + B + C)
(B) F = ($\overline{A}$ + $\overline{B}$ + $\overline{C}$)($\overline{A}$ + B + $\overline{C}$)(A + $\overline{B}$ + $\overline{C}$)
(C) F = (A + B + $\overline{C}$)(A + $\overline{B}$ + $\overline{C}$)($\overline{A}$ + B + $\overline{C}$)($\overline{A}$ + $\overline{B}$ + C)($\overline{A}$ + $\overline{B}$ + $\overline{C}$)
(D) F = ($\overline{A}$ + $\overline{B}$ + C)($\overline{A}$ + B + C)(A + $\overline{B}$ + C)(A + B + $\overline{C}$)(A + B + C)

Open solution

Q4. GATE ECE 2014 Set 1

The Boolean expression (X+Y)(X+$\overline{Y}$)+$\overline{(X\overline{Y}) + \overline{X}}$ simplifies to

(A) X
(B) Y
(C) XY
(D) X+Y

Open solution

Q5. GATE ECE 2014 Set 2

For an n - variable Boolean function maximum number of prime implicants is

(A) 2(n-1)
(B) n/2
(C) 2^n
(D) $2^{(n-1)}$

Open solution

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Q6. GATE ECE 2013

In the sum of products function f (x,y,z) = $\sum$ m (2,3,4,5), the prime implicants are

(A) $\overline{X}Y, X\overline{Y}$

(B) $\overline{X}Y, X\overline{YZ}, X\overline{Y}Z$

(C) $\overline{X}Y\overline{Z}, \overline{X}YZ, X\overline{Y}$

(D) $\overline{X}Y\overline{Z}, \overline{X}YZ, X\overline{YZ}, X\overline{Y}Z$

Open solution

Q7. GATE ECE 2007

The Boolean function Y=AB+CD is to be realized using only 2-input NAND gates. The minimum number of gates required is

(A) 2 (B) 3 (C) 4 (D) 5

Open solution

Q8. GATE ECE 2006

The number of product terms in the minimized sum-of-product expression obtained through the following k-map is (where , "d" denotes don't care states)

1	0	0	1
0	d	0	0
0	0	d	1
1	0	0	1

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(A) 2 (B) 3 (C) 4 (D) 5

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Q9. GATE ECE 2005

The Boolean expression for the truth table shown is

A	B	C	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Open solution

(A) B(A+C)($\overline{A}+\overline{C}$)

(B) $B\left(A+\overline{C}\right)\left(\overline{A}+C\right)$

(C) $\overline{B}$(A+$\overline{C}$)($\overline{A}$+C)

(D) $\overline{B}$(A+C)($\overline{A}+\overline{C}$)

Q10. GATE ECE 2003

The number of distinct Boolean expressions of 4 variables is

(A) 16 (B) 256 (C) 1024 (D) 65536

Open solution

Q11. GATE ECE 1999

The Logical expression $Y = A + \overline{A}B$ is equivalent to

(A) y = AB (B) y = $\overline{A}$ B (C) y = $\overline{A}$ + B (D) y = A + B

Open solution

Q12. GATE ECE 1998

Two 2' s complement numbers having sign bits x and y added and the sign bit of the result is z. Then, the occurrence of overflow is indicated by the Boolean function.

(A) xyz

(B) $\overline{x}\,\overline{y}\,\overline{z}$

(C) $\overline{x}\,\overline{y}$ z + x y $\overline{z}$

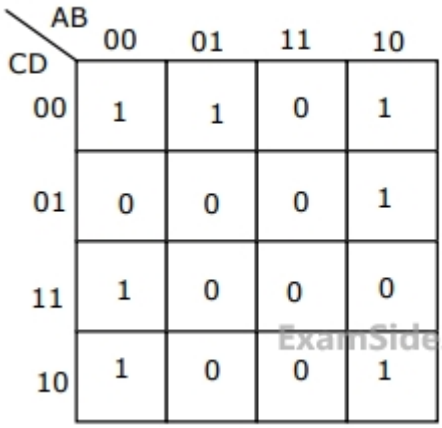
(D) xy + yz + zx

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Q13. GATE ECE 1998

The K-map for a Boolean function is shown in figure. The number of essential prime implicants for this function is



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(A) 4 (B) 5 (C) 6 (D) 8

Open solution

Q14. GATE ECE 1990

The number of Boolean functions that can be generated by n variable is equal to:

(A) 2^n (B) 2^{2^n} (C) 2^{n-1} (D) 2^{n+1}

Open solution

Marks 2 14 questions

Q15. GATE ECE 2026

Consider the four-variable Boolean function, $f(w,x,y,z) = \Sigma m(0,2,5,7,8,10,13,14,15)$ with ' w ' as MSB and ' z ' as LSB. Which of the following expressions is/are the valid form(s) of $f(w,x,y,z)$?

(A) $\bar{x}\bar{z} + xz + wxy$ (B) $xz + wxy + w\bar{x}\bar{z} + \bar{w}xy\bar{z}$

(C) $\bar{x}\bar{z} + wxy + w\bar{x}\bar{z} + \bar{w}xy\bar{z}$ (D) $\bar{x}\bar{z} + xz + wy\bar{z}$

Open solution

Q16. GATE ECE 2017 Set 1

Which one of the following gives the simplified sum of products expression for the Boolean function $F = m_0 + m_2 + m_3 + m_5$, where $F = m_0 + m_2 + m_3 + m_5$, are minterms corresponding to the inputs A, B and C with A as the MSB and C as the LSB?

(A) $\overline{A}B + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}\,C$ (B) $\overline{A}\,\overline{C} + \overline{A}B + A\overline{B}C$

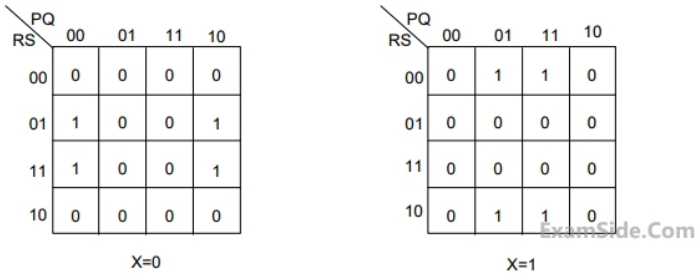
(C) $\overline{A}\,\overline{C} + A\,\overline{B} + A\overline{B}C$ (D) $\overline{A}\,BC + \overline{A}\,\overline{C} + A\,\overline{B}C$

Open solution

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Q17. GATE ECE 2016 Set 3

Following is the k-map of a Boolean function of five variables P, Q, R, S and X. The minimum sum-of-products expression for the function is



PQ / RS	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

X=0

PQ / RS	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	1	1	0

X=1

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(A) $\overline{P}\,\overline{Q}\,S\overline{X} + P\overline{Q}\,S\overline{X} + Q\,\overline{R}\,\overline{S}X + QR\overline{S}\,X$

(B) $\overline{Q}\,S\,\overline{X} + Q\,\overline{S}\,X$

(C) $\overline{Q}\,S\,X + Q\,\overline{S}\,\overline{X}$

(D) $\overline{Q}\,S + Q\,\overline{S}$

Open solution

Q18. GATE ECE 2015 Set 2

A function of Boolean variables X,Y and Z is expressed in terms of the min-terms as F(X, Y, Z)=$\sum$ m(1,2,5,6,7) Which one of the product of sums given below is equal to the funtion F(X, Y, Z)?

(A) $(\overline{X}+\overline{Y}+\overline{Z}).(\overline{X}+Y+Z).(X+\overline{Y}+\overline{Z})$

(B) $(X+Y+Z).(X+\overline{Y}+\overline{Z}).(\overline{X}+Y+Z)$

(C) $(\overline{X}+\overline{Y}+Z).(\overline{X}+Y+\overline{Z}).(X+\overline{Y}+Z).(X+Y+\overline{Z}).(X+Y+Z)$

(D) $(X+Y+\overline{Z}).(\overline{X}+Y+Z).(\overline{X}+Y+\overline{Z}).((\overline{X}+\overline{Y}+Z).(\overline{X}+\overline{Y}+\overline{Z})$

Open solution

Q19. GATE ECE 2015 Set 1

The Boolean expression F(X, Y, Z)= $\overline{X}Y\overline{Z} + X\overline{Y\,Z} + XY\overline{Z} + XYZ$ converted into the canonical product of sum (POS) form is

(A) $(x+y+z)(x+y+\overline{z})(x+\overline{y}+\overline{z})(\overline{x}+y+\overline{z})$

(B) $(x+\overline{y}+z)(\overline{x}+y+\overline{z})(\overline{x}+\overline{y}+z)(\overline{x}+\overline{y}+z)$

(C) $(x+y+z)(\overline{x}+y+\overline{z})(x+\overline{y}+z)(\overline{x}+\overline{y}+\overline{z})$

(D) $(x+\overline{y}+\overline{z})(\overline{x}+y+z)(\overline{x}+\overline{y}+z)(x+y+z)$

Open solution

Q20. GATE ECE 2014 Set 1

Consider the Boolean function, F(w,x,y,z)=wy+ xy +$\overline{w}\,xyz + \overline{w}\,\overline{x}y + xz + \overline{x}\,\overline{y}\,\overline{z}$ Which one of the following is the complete set of essential prime implicants?

(A) $w, y, xz, \overline{x}\,\overline{z}$

(B) w, y, xz

(C) $y, \overline{x}\,\overline{y}\,\overline{z}$

(D) y, xz, $\overline{x}\,\overline{z}$

Open solution

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Q21. GATE ECE 2009

If X=1 in the logic equation $\left[X+Z\left\{\overline{Y}+(\overline{Z}+X\overline{Y})\right\}\right]\left\{\overline{X}+\overline{Z}(X+Y)\right\}=1$, then

(A) Y=Z (B) Y= $\overline{Z}$ (C) Z= 1 (D) Z= 0

Open solution

Q22. GATE ECE 2007

The Boolean expression Y= $\overline{A}\,\overline{B}\,\overline{C}\,D+\overline{A}BC\overline{D}+A\overline{B}\,\overline{C}\,D+AB\overline{C}\,\overline{D}$ can be minimized to

(A) Y = $\overline{A}\,\overline{B}\,\overline{C}\,D+\overline{A}B\overline{C}+A\overline{C}D$

(B) Y = $\overline{A}\,\overline{B}\,\overline{C}\,D+BC\overline{D}+A\overline{B}\overline{C}\,D$

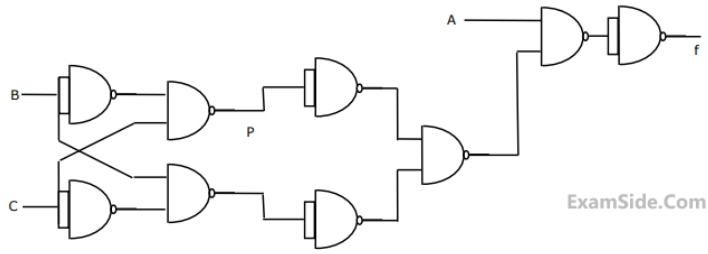
(C) Y=$\overline{A}\,BC\,\overline{D}+\overline{B}\,\overline{C}D+A\overline{B}\overline{C}\,D$

(D) Y= $\overline{A}\,BC\,\overline{D}+\overline{B}\,\overline{C}D+AB\overline{C}\,\overline{D}$

Open solution

Q23. GATE ECE 2006

The point p in the following figure is stuck- at-1. The output f will be



(A) $\overline{AB\overline{C}}$ (B) $\overline{A}$ (C) $AB\overline{C}$ (D) A

Open solution

Q24. GATE ECE 2004

The Boolean expression AC + B$\overline{C}$ is equivalent to

(A) $\overline{A}\,C+B\overline{C}+AC$ (B) $\overline{B}C+AC+B\overline{C}+\overline{A}C\overline{B}$

(C) AC+$B\overline{C}+\overline{B}C+ABC$ (D) $\overline{A}\,B\,\overline{C}+A\,B\,\overline{C}+A\overline{B}\,C+ABC$

Open solution

Q25. GATE ECE 2004

A Boolean function 'f' of two variables x and y is defined as follows: f(0,0)=f(0,1)=f(1,1)=1;f(1,0)=0 Assuming complements of x and y are not available, a minimum cost solution for realizing F using only 2-input NOR gates and 2-input OR gates (each having unit cost) would have a total cost

(A) 1 unit (B) 4 unit (C) 3 unit (D) 2 unit

Open solution

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Q26. GATE ECE 2003

Open solution

If the functions W, X, Y and Z are as follows
W= R+$\overline{P}Q + \overline{R}$ S
X = $PQ\overline{R}\,\overline{S} + \overline{P}\,\overline{Q}\,\overline{R}\,\overline{S} + P\overline{Q}\,\overline{R}\,\overline{S}$
Y = $\overline{RS + OR + P\overline{Q} + \overline{PQ}}$
Z = $\overline{R + S + PQ + \overline{P}\,\overline{Q}\,\overline{R} + P\overline{Q}\,\overline{S}}$

(A) W=Z, X=$\overline{Z}$ (B) W=$\overline{Z}$, X=Y (C) W=Z, X=Y (D) W=Y= $\overline{Z}$

Q27. GATE ECE 1999

Open solution

The minimized form of the logical expression $(\overline{A}\,\overline{B}\,\overline{C} + B\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + AB\overline{C})$

(A) $\overline{AC} + B\overline{C} + \overline{A}B$ (B) $A\overline{C} + \overline{B}C + \overline{A}B$
(C) $\overline{A}C + \overline{B}C + \overline{A}B$ (D) $A\overline{C} + \overline{B}C + A\overline{B}$

Q28. GATE ECE 1999

Open solution

For a binary half-subtractor having two inputs A and B, the correct set of Logical expressions for the output D(=Aminus B) and X(=Borrow) are

(A) D = AB + $\overline{A}B, X = \overline{A}B$ (B) $D = \overline{A}B + A\overline{B} + A\overline{B}, X = A\overline{B}$
(C) $D = \overline{A}B + A\overline{B}, X = \overline{A}B$ (D) $D = AB + \overline{A}\,\overline{B}, X = A\overline{B}$

Marks 4 1 questions

Q29. GATE ECE 1996

Open solution

Given the Boolean function F in three variables R, S and T as F= $\overline{R}\,S\overline{T} + R\overline{S}\,T + RST$
(a) Express F in the minimum sum-of-products from.
(b) Express F in the minimum product-of-sums from.
(c) Assuming that both true and complement froms of the input variables are avialable, draw a cicruit to implement F using the minimum number of 2input NAND gates only.

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Marks 5 2 questions

Q30. GATE ECE 1999

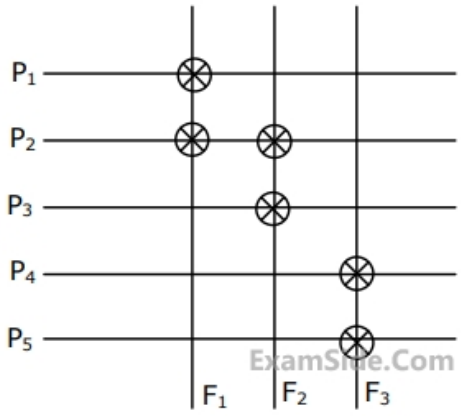
In certain application, four inputs A, B, C, D (both true and complement forms available)are fed to logic circuit, producing an output F which operates a relay. The relay turns on when F(ABCD)=1 for the following states of the inputs (ABCD):'0000', '0010' ,0101',0110','1101' and '1110'. States '1000' and '1001' do not occur, and for the remaining states, the relay is off. Minaining states, the relay is off. Minimize F with the help of a Karnaugh map and realize it using a minimum number of 3- input NAND gates.

Open solution

Q31. GATE ECE 1996

Its is desired to generate the following three Boolean functions. $F_1 = a\,\overline{b}\,c + \overline{a}\,b\overline{c} + bc$
$F_2 = a\,\overline{b}\,c + ab + \overline{a}\,b\overline{c}$
$F_3 = \overline{a}\,\overline{b}\,\overline{c} + abc + \overline{a}c$

By using an OR gate array as shown in figure where P_1 to P_5 are the product terms in one or more of the variables a, $\overline{a}$, b, $\overline{b}$, c and $\overline{c}$.



Open solution

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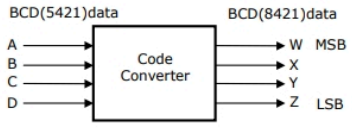
Marks 10 2 questions

Q32. GATE ECE 1995

A 'code converter' is to be designed to convert from the BCD(5421) to the normal BCD (8421). The input BCD combinations for each digit are below. A block diagram of the converter is shown in figure.

Open solution

Decimal	BCD(5421)			
	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	1	0	0	0
6	1	0	0	1
7	1	0	1	0
8	1	0	1	1
9	1	1	0	0



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(A) draw K-map for outputs, W, X, Y, and Z.
(B) Obtain minimized expression for the outputs W, X, Y, and Z.

Q33. GATE ECE 1991

The four variable function f is given in terms of min-terms as: f (A B C D) = $\sum$ m (2,3,8,10,11,12,14,15 .) Using the K-map minimize the function in the sum of products form. Also, given the realization using only two-input NAND gates .

Open solution

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Answer Key

Marks 1

1. **A**	2. **B, C**	3. **C**	4. **A**	5. **B**
6. **A**	7. **B**	8. **A**	9. **A**	10. **D**
11. **D**	12. **C**	13. **A**	14. **B**	

Marks 2

15. **A, D**	16. **B**	17. **B**	18. **B**	19. **A**
20. **D**	21. **D**	22. **D**	23. **D**	24. **D**
25. **D**	26. **A**	27. **A**	28. **C**	

Marks 4

29. **(a) F =RT+** $\overline{R}\,S\,T$ **(b) F** **=(R+**$\overline{T}$$)(\overline{R}+T)(R+S)$	

Marks 5

30. **F =**$\overline{C}$ **D+BC**$\overline{D}$ **+** $\overline{A}\,\overline{B}\,\overline{D}$	31. P_1 **=** $\overline{a}$ **b** P_2 **= ac** P_3 **= b**$\overline{c}$ **IF** P_4 **= bc then** P_5 **=** $\overline{a}\,\overline{c}$ **IF** P_4 **=** $\overline{a}\,\overline{c}$ **then IF** P_5 **=bc**

Marks 10

32. **(b) W= AB+ACD** **X=**$\overline{A}B+A\overline{C}D+A\,\overline{B}\,\overline{D}$ **or** $X=\overline{A}\,B+A\overline{B}\overline{C}+AC\overline{D}$ **OR** **X=** $\overline{A}B+A\,\overline{B}\,\overline{C}+A\,\overline{B}\,\overline{D}$ **Y=**$\overline{A}C+A\overline{C}D+AC\overline{D}$ **Z=**$\overline{A}\,D+A\,\overline{D}$	33. **f = (A, B, C, D)=A**$\overline{D}+AC+\overline{B}C$

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